



6567 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
G20.0+0.0				
	1	G20.0+0.0 (ABC)	MIRI Medium Resolution Spectroscopy	(1) G20.0+0.0
G20.0+0.6				
	2	G20.0+0.6 (ABC)	MIRI Medium Resolution Spectroscopy	(2) G20.0+0.6
Background				
	3	Background	MIRI Medium Resolution Spectroscopy	(3) Background

ABSTRACT

NRAO Proposal ID: GBT/24B-101

The Warm Ionized Medium (WIM) consists of diffuse ionized gas in the Galactic mid-plane and is the last major component of the interstellar medium to be studied at high angular and spectral resolution. OB-type stars, with a scale height of 30pc, are thought to be the main source of ionization. But hot low-mass evolved stars (UV stars), with a scale height of 800pc, may significantly contribute to the WIM. Here, we investigate the importance of UV stars by determining the hardness of the radiation field as a function of Galactic latitude. To do this we request observations of

JWST Proposal 6567 (Created: Wednesday, April 2, 2025, 4:00:22PM Eastern Standard Time) - Overview

radio recombination lines (RRLs) with the GBT and forbidden lines of Ne, Ar, and S with the JWST at two Galactic latitudes, G20.0+0.0 and G20.0+0.6. These data will be used to constrain the radiation field using CLOUDY simulations that will allow us to evaluate the importance of UV stars.

OBSERVING DESCRIPTION

N/A

Proposal 6567 - Targets - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	G20.0+0.0	RA: 18 27 31.7789 (276.8824121d) Dec: -11 29 18.98 (-11.48861d) Equinox: J2000	Proper Motion RA: 0.0 mas/yr Proper Motion Dec: 0.0 mas/yr Parallax: 0.0" Epoch of Position: 2000.0	
	Comments: GBT Position Category=ISM Description=[Warm ionized medium] Extended=YES				
	(2)	G20.0+0.6	RA: 18 25 21.8391 (276.3409962d) Dec: -11 12 32.13 (-11.20893d) Equinox: J2000	Proper Motion RA: 0.0 mas/yr Proper Motion Dec: 0.0 mas/yr Parallax: 0.0" Epoch of Position: 2000.0	
	Comments: Category=ISM Description=[Warm ionized medium] Extended=YES				
	(3)	Background	RA: 18 30 11.8142 (277.5492258d) Dec: -11 55 58.63 (-11.93295d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				

Proposal 6567 - Observation 1 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Observation	Proposal 6567, Observation 1: G20.0+0.0 (ABC)												Wed Apr 02 21:00:22 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Background (Obs 3)]													
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	G20.0+0.0	RA: 18 27 31.7789 (276.8824121d) Dec: -11 29 18.98 (-11.48861d) Equinox: J2000				Proper Motion RA: 0.0 mas/yr Proper Motion Dec: 0.0 mas/yr Parallax: 0.0" Epoch of Position: 2000.0							
	Comments: GBT Position													
	Category=ISM													
	Description=[Warm ionized medium]													
Extended=YES														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	1	LONG(C)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	2	MEDIUM(B)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	2	MEDIUM(B)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	3	SHORT(A)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215	185559	
	3	SHORT(A)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215	185559	

Proposal 6567 - Observation 1 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
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Proposal 6567 - Observation 2 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Observation	Proposal 6567, Observation 2: G20.0+0.6 (ABC)												Wed Apr 02 21:00:22 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Background (Obs 3)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	G20.0+0.6	RA: 18 25 21.8391 (276.3409962d)				Proper Motion RA: 0.0 mas/yr							
	Dec: -11 12 32.13 (-11.20893d)				Proper Motion Dec: 0.0 mas/yr									
	Equinox: J2000				Parallax: 0.0"									
					Epoch of Position: 2000.0									
Comments: Category=ISM Description=[Warm ionized medium] Extended=YES														
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	1	LONG(C)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	2	MEDIUM(B)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	2	MEDIUM(B)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215		
	3	SHORT(A)	MRSLONG		FASTR1	92	1	1	Dither 1	4	4	1021.215	185559	
	3	SHORT(A)	MRSSHORT		FASTR1	92	1	1	Dither 1	4	4	1021.215	185559	

Proposal 6567 - Observation 2 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
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Proposal 6567 - Observation 3 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Observation	Proposal 6567, Observation 3: Background												Wed Apr 02 21:00:22 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [G20.0+0.0 (ABC) (Obs 1), G20.0+0.6 (ABC) (Obs 2)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(3)	Background	RA: 18 30 11.8142 (277.5492258d) Dec: -11 55 58.63 (-11.93295d) Equinox: J2000					Epoch of Position: 2000					
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	LONG(C)	MRSLONG		FASTR1	92	1	1	Dither 1	2	2	510.607	
	1	LONG(C)	MRSSHORT		FASTR1	92	1	1	Dither 1	2	2	510.607	
	2	MEDIUM(B)	MRSLONG		FASTR1	92	1	1	Dither 1	2	2	510.607	
	2	MEDIUM(B)	MRSSHORT		FASTR1	92	1	1	Dither 1	2	2	510.607	
	3	SHORT(A)	MRSLONG		FASTR1	92	1	1	Dither 1	2	2	510.607	185559
	3	SHORT(A)	MRSSHORT		FASTR1	92	1	1	Dither 1	2	2	510.607	185559

Proposal 6567 - Observation 3 - Ionization State of the Warm Ionized Medium in the Inner Galaxy

Special Requirements	Sequence Observations 3, 1, 2 (reordered), Non-interruptible
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